Lab 47 — Malware Symptom Response and Browser Problem Resolution

> **Course:** CompTIA Certified A+ Training (Core 1 and Core 2) (TGS-2024048317)
> **Topic 08:** Software Troubleshooting — Core 2, 23% of Core 2
> **Exam objective:** Recognise malware and browser symptoms, apply the containment sequence and resolve browser-specific problems (Core 2 objectives 3.2 and 3.3).

Goal

Browser symptoms are how most malware infections are first reported, and the containment sequence matters more than the cleanup: a machine left on the network while you investigate keeps spreading. You build both the symptom map and the response sequence.

What you'll produce

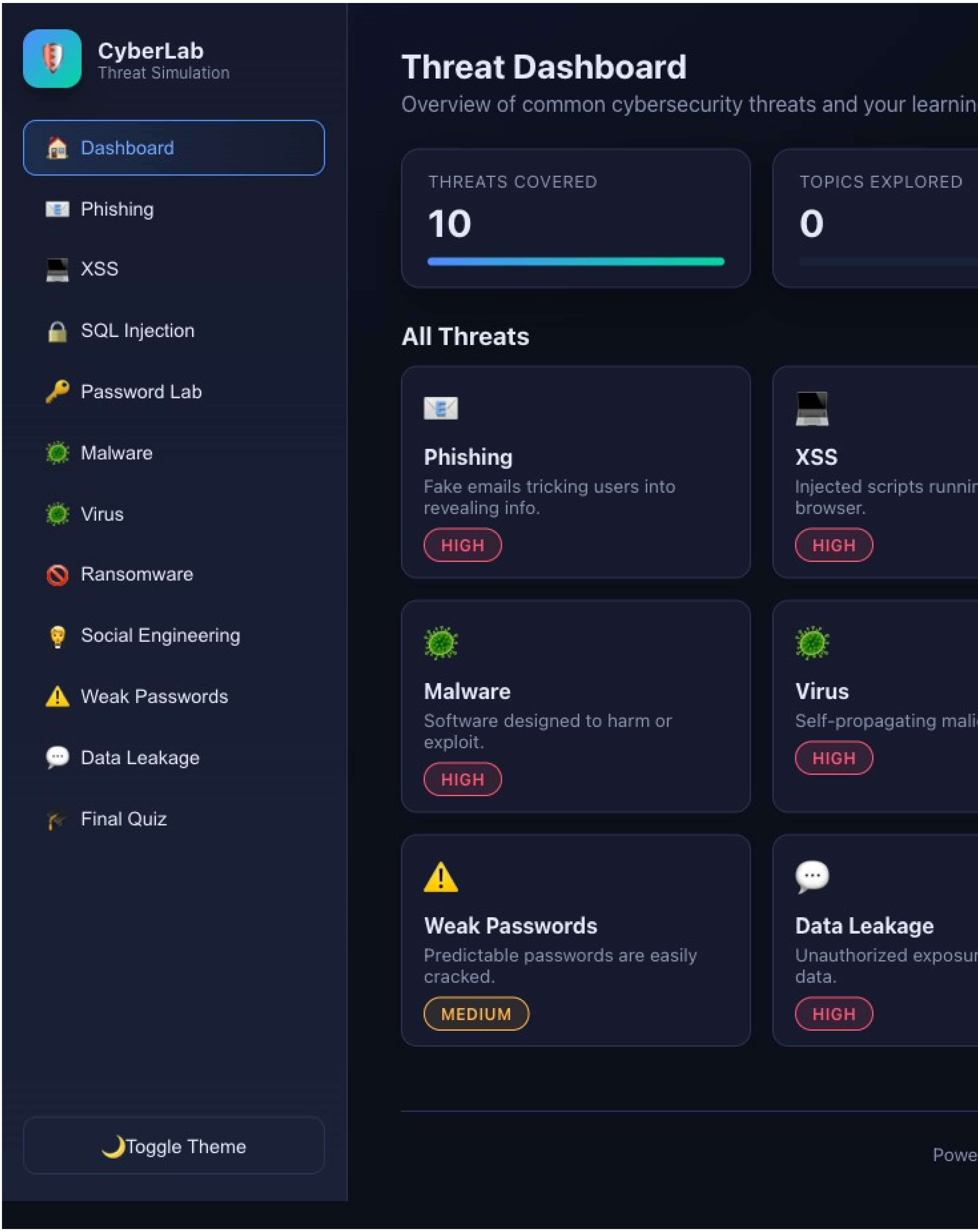
A malware and browser symptom map with the response sequence and the containment decision point for each.

Tools and equipment

Cybersecurity Simulator, Windows PC, browser settings, Windows Security

Browser tools used in this lab

- **Cybersecurity Simulator** — https://alfredang.github.io/cybersecuritysimulator/



Cybersecurity Simulator — the panels and fields this lab uses.

Safety

- Disconnect mains power and hold the power button for 15 seconds before working inside any machine.
- Wear an anti-static strap connected to an earthed point; hold boards and cards by their edges.
- **Never open a power supply unit or a CRT monitor** — both retain a lethal charge after being unplugged.
- Never puncture, compress or charge a swollen lithium battery; isolate it and follow hazardous disposal procedure.

Step-by-step

1. Record the containment rule that governs everything else: at the first credible sign of malware, disconnect the machine from the network before beginning any investigation.
2. Build the malware symptom list: unexpected pop-ups, browser redirection, certificate warnings, sudden slowdown, disabled security tools, unexplained network traffic, new unfamiliar applications and files that will not open.
3. Record the false alert symptom: a message claiming antivirus is missing or expired, delivered by a web page rather than by the installed product, is itself a social engineering attack.
4. Diagnose certificate warnings by separating the causes: an expired certificate, a self-signed certificate, a wrong system clock, or an on-path interception — and record the check for each.
5. Record that a certificate warning on a major site with a correct system clock is a serious signal and the connection must not be continued.
6. Diagnose browser redirection: check the configured search provider and homepage, review installed extensions, check the proxy settings, and inspect the browser shortcut target for appended arguments.
7. Record why removing a malicious extension from the browser is not enough on its own — if a dropper persists in Windows it simply reinstalls the extension.
8. Review the browser security settings that matter: pop-up blocker, clearing browsing data and cache, private browsing mode, ad blockers and sign-in data synchronisation.
9. Record the browser installation rule: download only from the vendor's own site, verify the hash where published, and treat extensions from outside the official store as untrusted.
10. Apply the seven-step removal procedure to a browser-symptom scenario and state the specific action at each step.
11. Record the post-cleanup verification: confirm the symptom is gone, confirm security tools are running and updated, check for persistence in startup items and scheduled tasks, and rescan.
12. Record the final step that is most often skipped: educate the user on how the infection arrived, because the same user on the same machine will otherwise repeat it.

Test it — verification

Your map covers at least eight malware symptoms and four browser symptoms with a response for each; network disconnection appears as the first containment action; the certificate warning causes are correctly separated; and post-cleanup verification includes checking persistence mechanisms.

Troubleshooting this lab

Symptom	What to check
A command returns "command not found"	Re-run the `apt-get install` step at the start of the lab — the playground starts with a minimal package set.
The Killercoda terminal has reset	The playground times out when idle. Reopen it and re-run the setup commands from step 1.
A browser tool will not load	Check the URL against `labs/tools.md`. All four tools run entirely client-side and need no login.
Output differs from the guide	Record what you actually observed — your environment differs from the reference, and explaining the difference is part of the exercise.

Review questions

1. State the exam objective this lab maps to, in your own words.
2. Which single step in this lab would you perform first on a real support call, and why?
3. What evidence would you attach to a support ticket to show this work was completed correctly?
4. Name one thing that would make this procedure fail, and how you would recognise it.

Record your evidence

Complete worksheet.md as you work through this lab and keep it — the Practical Performance assessment mirrors these tasks.

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